

OSSP

Practical-Assignment1

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1.

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[20];

    printf("Enter command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

```
haadi_hasil@DESKTOP-J6ARPCX:~$ nano practical1.c
haadi_hasil@DESKTOP-J6ARPCX:~$ gcc practical1.c
haadi_hasil@DESKTOP-J6ARPCX:~$ ./a.out
Enter command: ls
Parent PID: 659
Child PID: 660
a.out a8 b1 copy.c f1.c fork3.c gadilafile newfile1 permission_lab pipe6.c practical1.c readwrite readwrite.c.save
a5 ac b2 dir65 fork.c fork4.c linux_task4 open2.c pipe1.c pipe7.c read1.c readwrite.c
Child completed
haadi_hasil@DESKTOP-J6ARPCX:~$ |
```

2.

```
L1i:                256 KiB (8 instances)
L2:                 10 MiB (8 instances)
L3:                 20 MiB (1 instance)
NUMA:
  NUMA node(s):      1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:          Not affected
  Indirect target selection: Not affected
  Itlb multihit:       Not affected
  L1tf:                Not affected
  Mds:                 Not affected
  Meltdown:           Not affected
  Mmio stale data:     Not affected
  Old microcode:       Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:            Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:   Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:          Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:          Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:              Not affected
  Tsa:                Not affected
  Tsx async abort:    Not affected
  Vmscape:            Not affected
haadi_hasil@DESKTOP-J6ARPCK:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro/
haadi_hasil@DESKTOP-J6ARPCK:~$ ps
  PID TTY          TIME CMD
   341 pts/0    00:00:00 bash
   727 pts/0    00:00:00 ps
```

```

haadi_hasil@DESKTOP-J6ARPCK:~$ uname
Linux
haadi_hasil@DESKTOP-J6ARPCK:~$ lscpu
Architecture:          x86_64
  CPU op-mode(s):      32-bit, 64-bit
  Address sizes:        39 bits physical, 48 bits virtual
  Byte Order:           Little Endian
CPU(s):                 16
  On-line CPU(s) list: 0-15
Vendor ID:              GenuineIntel
  Model name:           13th Gen Intel(R) Core(TM) i5-13450HX
  CPU family:           6
  Model:                183
  Thread(s) per core:   2
  Core(s) per socket:   8
  Socket(s):            1
  Stepping:             1
  BogoMIPS:             5222.40
  Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                        r pge mca cmov pat pse36 clflush mmx fxsr sse s
                        se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                        _tsc rep_good nopl xtopology tsc_reliable nonst
                        op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                        sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                        opcnt tsc_deadline_timer aes xsave avx f16c rdr
                        and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                        brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                        d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                        2 erms invpcid rdseed adx smap clflushopt clwb
                        sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                        vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                        vdiri movdir64b fsrm md_clear serialize ibt flu
                        sh_llld arch_capabilities

Virtualization features:
  Virtualization:       VT-x
  Hypervisor vendor:    Microsoft
  Virtualization type:  full
Caches (sum of all):
  L1d:                  384 KiB (8 instances)
  L1i:                  256 KiB (8 instances)
  L2:                   10 MiB (8 instances)

```

- Linux OS acts as a link between computer hardware and applications.
- Linux manages the CPU using process scheduling.
- Linux manages storage using files, directories, and file systems.
- `uname` is used to check the Linux kernel and system information.
- `lscpu` gives details about the CPU, cores, threads, and architecture.
- `lsblk` shows the storage devices, disks, SSDs, and partitions.
- `ps` is used to see the running processes and their PIDs.
- `top` shows the real-time CPU, memory, and process usage.